Beamforming with Environment Sensing for WET

Ana Cecília Silveira Fernandes, Isabel Mendes

I. Proposal

Wireless Energy Transfer via Radio Frequency (RF-WET) is a promising solution to meet the growing energy demands of Internet of Things (IoT) devices. This technology aims to reduce dependence on conventional batteries, which present limitations in terms of durability, maintenance costs, and environmental impacts caused by improper disposal [1]. The fundamental principle of RF-WET lies in the conversion of electromagnetic energy, emitted by a transmitting source, into usable electrical energy by receiving devices equipped with rectifying circuits (rectennas). This architecture enables continuous energy supply, allowing uninterrupted operation of sensors, actuators, and other low-power devices, thereby fostering the consolidation of sustainable IoT ecosystems. However, the development of this technology still faces significant technical challenges. Energy transfer efficiency is affected by propagation losses, undesired absorption, and regulatory constraints on RF emissions [2]. Moreover, the requirement for short-range operation and precise alignment between transmitter and receiver limits its applicability in large-scale and complex scenarios. In this context, the present project aims to maximize the energy harvested by IoT devices while ensuring the safety of the RF-WET system for living beings. To achieve this, two complementary strategies are proposed: (i) environmental sensing for human detection and (ii) beamforming optimization at the transmitter (Power Beacon – PB). The methodology will be structured into three main stages:

1) Dataset generation: a synthetic dataset of virtual scenarios will be created using a professional open-source software for 3D modeling and simulation. At this stage, the inclusion of only humans will be considered.
2) Implementation of a machine learning (ML) model for human detection: based on the simulated scenarios, a computer vision model will be trained to identify the presence and position of humans within the scenes.
3) Beamforming optimization supported by ML: leveraging the detected human positions and the prior knowledge of IoT device locations, an ML-based model will be developed to design beamforming strategies at the transmitter.

This integrated approach has the potential to significantly advance wireless energy transfer, contributing to the development of more autonomous, safe, and sustainable IoT networks.

II. Methodology

A. First Part

The *Blensor software* [3], [4] is a professional, open-source, and completely free application widely adopted in the digital content creation industry. It provides a comprehensive set of tools for 3D modeling, animation, rendering, physics-based simulation, video editing, and even game development.

In this study, Blensor will be employed to generate a synthetic database of virtual scenarios in which humans can be placed. These scenarios may include both indoor environments (e.g., rooms, offices, hallways) and outdoor environments (e.g., streets, squares), thereby enabling the creation of diverse and realistic datasets. Such datasets will serve as the training foundation for deep learning models.

Once the dataset has been sufficiently constructed, the next phase will focus on training deep learning models to detect the presence of humans within the generated scenes.

Among the object detection models considered, the following stand out:

- YOLO (You Only Look Once) [5]: a state-of-the-art algorithm designed for real-time object detection, known for its balance between speed and accuracy.
- Faster R-CNN [6]: a robust detection model that combines high precision with flexibility, extensively applied in both academic and industrial contexts.
- Apple Create ML [7]: a machine learning framework integrated into the macOS ecosystem, which facilitates the training of computer vision models and their seamless deployment into iOS and macOS applications.

In summary, the proposed pipeline consists of three stages: (1) synthetic dataset generation using Blensor, (2) training of deep learning models, and (3) deployment of these models for human detection in both virtual and real-world scenarios.

B. Second Part

After the human detection stage, the methodology advances toward the optimization of beamforming strategies at the transmitter. The objective is to design transmission weights for the antenna array that maximize the harvested energy at IoT devices while respecting safety constraints related to human exposure. The simulation environment will combine information from the detection module with the spatial distribution of IoT devices. Indoor conditions will be considered, including realistic channel models with path loss, fading, and multipath propagation. The energy conversion process will follow a non-linear harvesting model, and compliance with regulatory exposure limits will be evaluated to ensure safe operation in the presence of humans [8].

From the generated scenarios, a labeled database will be constructed in which each entry consists of the positions of humans and devices together with the corresponding optimal beamforming solution. This database will serve as the foundation for training a machine learning model capable of approximating optimal solutions in real time. At this stage, the choice of technique remains open, with lightweight neural networks, compact attention-based architectures, and hybrid approaches all under consideration. The selection will ultimately depend on trade-offs among accuracy, computational complexity, and feasibility of integration in real systems.

The performance of the proposed model will be assessed in terms of harvested energy per device, frequency of safety violations, robustness to uncertainties in detection, and comparison with classical baselines such as Equal-Gain beamforming [9]. Once validated, the system will integrate all components into an end-to-end pipeline that combines human detection, information fusion, beamforming computation, safety verification, and continuous monitoring of efficiency and compliance.

III. Results

References

[1] O. L. López, H. Alves, R. D. Souza, S. Montejo-Sánchez, E. M. G. Fernández, and M. Latva-Aho, "Massive wireless energy transfer: Enabling sustainable iot toward 6g era," *IEEE Internet of Things Journal*, vol. 8, no. 11, pp. 8816–8835, 2021.

[2] X. Lu, P. Wang, D. Niyato, D. I. Kim, and Z. Han, "Wireless networks with rf energy harvesting: A contemporary survey," *IEEE Communications Surveys Tutorials*, vol. 17, no. 2, pp. 757–789, 2015.

[3] Blensor, "Free open source simulation package for light detection and ranging (lidar/ladar) and kinect sensors," acess in August, 2025. [Online]. Available: https://www.blensor.org

[4] S. Reitmann, L. Neumann, and B. Jung, "Blainder—a blender ai add-on for generation of semantically labeled depth-sensing data," *Sensors*, vol. 21, no. 6, p. 2144, 2021.

[5] Ultralytics, "Object detection," 2023, acess in August, 2025. [Online]. Available: https://docs.ultralytics.com/tasks/detect/

[6] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *Advances in neural information processing systems*, vol. 28, 2015.

[7] Apple, "Mlobjectdetector," acess in August, 2025. [Online]. Available: https://developer.apple.com/documentation/createml/mlobjectdetector

[8] E. Boshkovska, A. Koelpin, D. W. K. Ng, N. Zlatanov, and R. Schober, "Robust beamforming for swipt systems with non-linear energy harvesting model," in *IEEE SPAWC*, 2016.

[9] E. Ramadan, R. Morsi, E. Boshkovska, and R. Schober, "On the performance of wireless powered communication with non-linear energy harvesting," *arXiv preprint arXiv:1700.xxxx*, 2017.